KUSH AGARWAL

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SUMMARY

B.Tech Computer Science (AI) undergraduate, 2027 batch, who builds AI-enabled backend systems end to end in Python, FastAPI, React and PostgreSQL: design, testing, Dockerized deployment and iteration. Integrates LLMs, embeddings and vector retrieval (RAG-style) into products with guardrails, and treats every model output, including AI-assisted code from Claude, as a draft that is validated and unit-tested before it ships. Strong in data structures, graph algorithms, operating systems and networks, with 540 LeetCode problems solved at a 1938 contest rating. Seeking an R&D internship to research, prototype and harden AI-driven software.

EDUCATION

Noida Institute of Engineering & Technology 2023 – 2027
B.Tech, Computer Science & Engineering (Artificial Intelligence) — CGPA 8.66
CBSE Class XII — 86.20% | CBSE Class X — 94.00%

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL
AI / LLM: LLM APIs (OpenAI, Anthropic, Groq), prompt design and output validation, embeddings and FAISS vector retrieval (RAG), OCR and semantic analysis, Scikit-learn
Backend & APIs: FastAPI, Node.js / Express.js, REST API design and versioning, SQLAlchemy, Pydantic
Frontend: React.js, TypeScript, Streamlit
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, Docker, AWS (EC2, S3), Git/GitHub
Testing: pytest, property-based testing (Hypothesis), unit and integration testing
Core CS: Data Structures & Algorithms, Graph Algorithms, OOP & Object-Oriented Design, Operating Systems, Computer Networks, DBMS

PROJECTS

TaskFlow Pro — DAG Workflow & Scheduling Engine [GitHub] *Python, FastAPI, PostgreSQL, React, TypeScript, LLM APIs*

- Built a Kanban workflow platform for the Contata NCR Hackathon 2026 on a DAG engine that rejects circular dependencies with the exact loop, derives Blocked/Ready status and re-blocks dependents when a finished task regresses.
- Designed max-based topological schedule propagation that never double-counts delays on converging paths; pushes a delay through 1,000 tasks and 3,000 links in under 1 ms.
- Added LLM dependency suggestions behind five validation gates (schema, id check, de-duplication, quoted-evidence grounding, cycle dry run), with human approval and an offline fallback.
- Wrote 61 backend tests, including property-based and concurrency tests, passing on SQLite and PostgreSQL; built with Claude as a disclosed AI assistant.

AI-Powered Exam Generator & Evaluator *Python, Node.js, Streamlit, Groq API (LLM), OCR*

- Shipped an LLM-powered assessment platform that generates personalized exams from uploaded course content, taking it from idea through build and iteration.
- Built automated evaluation pipelines for handwritten, descriptive and coding answers using OCR and semantic analysis, with unit-tested REST services that validate LLM output rather than trusting it by default.

AI-Powered Log Analyzer & Root-Cause Analysis Tool *Python, FastAPI, PostgreSQL, Redis, FAISS, Docker*

- Engineered a Dockerized Redis Streams log-ingestion pipeline that scales horizontally across consumer groups for high-volume production log traffic.
- Automated root-cause classification with a unit-tested parsing pipeline and an embedding-backed FAISS similarity search that surfaces past incidents and fixes instantly.

ARAIA — Portfolio Diversification Under Macroeconomic Stress [Live Demo] *Python, FastAPI, React, AWS, FAISS*

- Built versioned, unit-tested FastAPI APIs for a React risk-analytics platform deployed on AWS (EC2 + S3); a FAISS-powered recommender cut simulated drawdown by 59%, validated by back-testing across stress scenarios.

EXPERIENCE

Python Developer Intern — LetsGrowMore Jul 2024 – Aug 2024

- Partnered with product stakeholders to scope feature requirements and translate business needs into technical specs, delivering through code review and iterative sprints.
- Built Python automation scripts (Pandas, Flask) that streamlined recurring data-processing workflows and removed repetitive manual effort for the team.

ACHIEVEMENTS & CERTIFICATIONS

- Shortlisted for the build round of the Contata NCR Hackathon 2026 (R&D Internship track).
- 540 DSA problems solved on LeetCode with a 1938 contest rating (Knight).
- Top 10 of 120+ teams, HackO'Clock national-level hackathon — built and demoed a working prototype under deadline.
- Cisco Networking Academy: Introduction to Modern AI | Infosys Springboard: Generative AI, Deep Learning, Python, React.js.